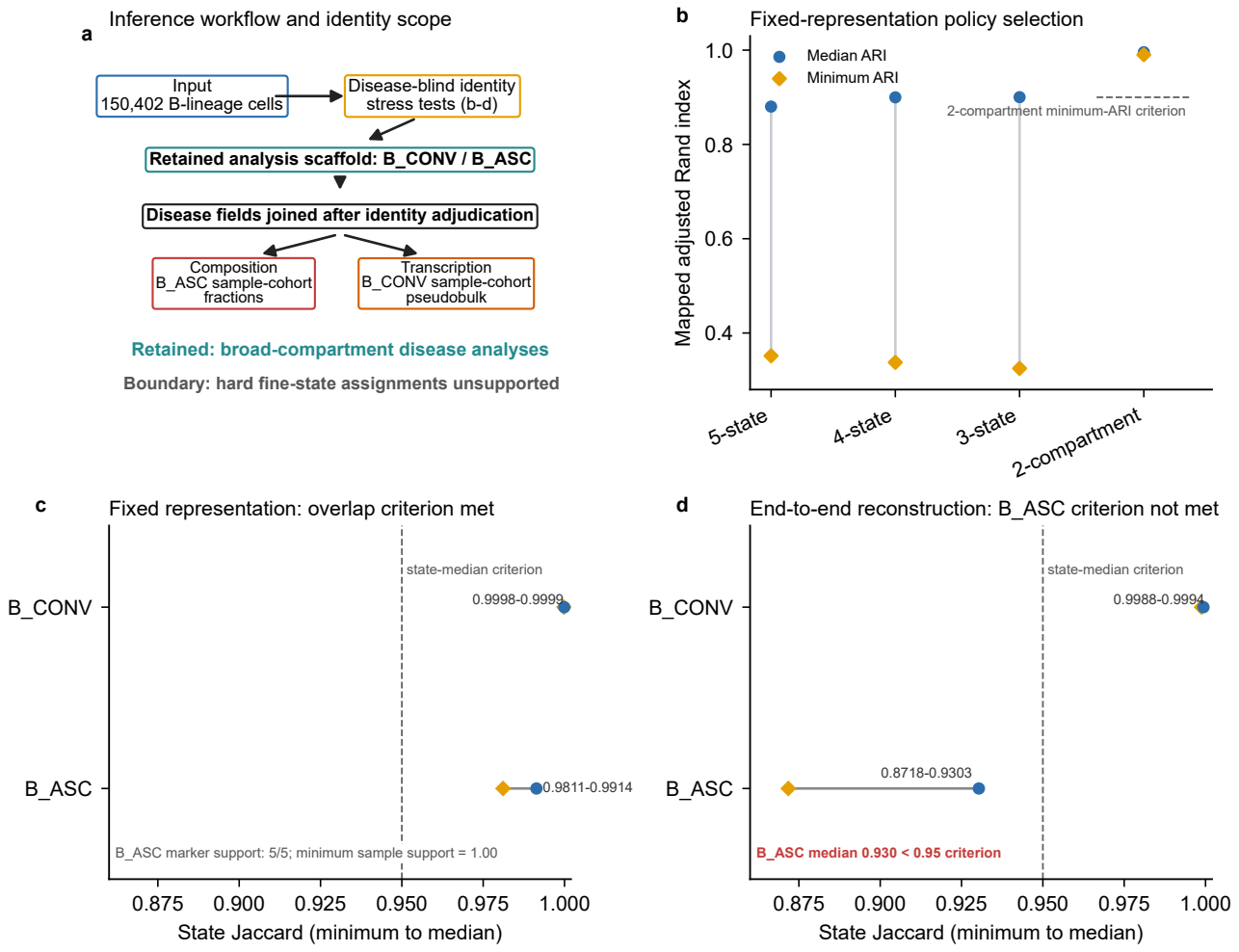

a
Inference workflow and identity scope
Input
150,402 B-lineage cells
Disease-blind identity stress tests (b-d)
Retained analysis scaffold: B_CONV / B_ASC
Disease fields joined after identity adjudication
Composition
B_ASC sample-cohort fractions
Transcription
B_CONV sample-cohort pseudobulk
Retained: broad-compartment disease analyses
Boundary: hard fine-state assignments unsupported
b
Fixed-representation policy selection
Median ARI
Minimum ARI
2-compartment minimum-ARI criterion
Mapped adjusted Rand index
1.0
0.8
0.6
0.4
5-state
4-state
3-state
2-compartment
c
Fixed representation: overlap criterion met
state-median criterion
B_CONV
0.9998-0.9999
B_ASC
0.9811-0.9914
B_ASC marker support: 5/5; minimum sample support = 1.00
0.875
0.900
0.925
0.950
0.975
1.000
State Jaccard (minimum to median)
d
End-to-end reconstruction: B_ASC criterion not met
state-median criterion
B_CONV
0.9988-0.9994
B_ASC
0.8718-0.9303
B_ASC median 0.930 < 0.95 criterion
0.875
0.900
0.925
0.950
0.975
1.000
State Jaccard (minimum to median)